

The Orchard Program

A Deep-Time Architecture for Interstellar Communication

Consolidated Technical Paper — Beacon Physics, Phase-1 Engineering, the Solar Gravitational Lens, and Self-Replicating Networks

Abstract

Classical SETI/METI implicitly assumes two civilizations transmit, listen, and stay interested simultaneously. Across galactic timescales this assumption almost never holds. This paper develops an alternative: replace *communication between civilizations* with *communication between epochs*. We derive the link-budget physics that favors narrow, repeated, multi-band beams over isotropic broadcasting; specify a buildable Phase-1 beacon (\$70M–\$250M) reachable with present technology; analyze the solar gravitational lens as a natural ~ 25 -km antenna beyond 550 AU; and model self-replicating ‘seed’ probes that fill the galaxy in $\sim 10^7$ – 10^8 years and survive deep time through redundancy. Eight figures and supporting tables quantify detection range, required transmitter power, antenna geometry, lens gain, propagation speed, and network survival. The deepest function of the architecture is not contact but **continuity**: a distributed, self-repairing memory of conscious life.

Whitepaper suite v1.0 · All cost and physics figures are order-of-magnitude engineering estimates anchored to real programs (Arecibo, Breakthrough Listen, SKA, FAST, Voyager) and the published SGL literature.

1. Introduction: The Temporal Isolation Problem

The Milky Way is ~13.6 billion years old. A technological civilization may persist for centuries, millennia, or—optimistically—millions of years. Even a million-year civilization occupies under 0.01% of galactic history. If two civilizations' lifetimes L are randomly placed within a communicative window W , the probability that they overlap in time is approximately:

$$P_{\text{overlap}} \approx \frac{2L}{W} \quad (L \ll W)$$

For $L = 10^4$ yr and $W = 10^{10}$ yr, $P_{\text{overlap}} \approx 2 \times 10^{-6}$. Two civilizations sharing the galaxy are overwhelmingly likely to miss each other in time: a signal sent today may arrive after its sender is extinct, and any reply after its receiver is gone. Real-time dialogue is therefore the wrong design target. The right target is **persistence and propagation**—infrastructure that does not require the founders to survive. Figure 1 fixes the basic constraint: radio travels at c , so travel time in years equals distance in light-years.

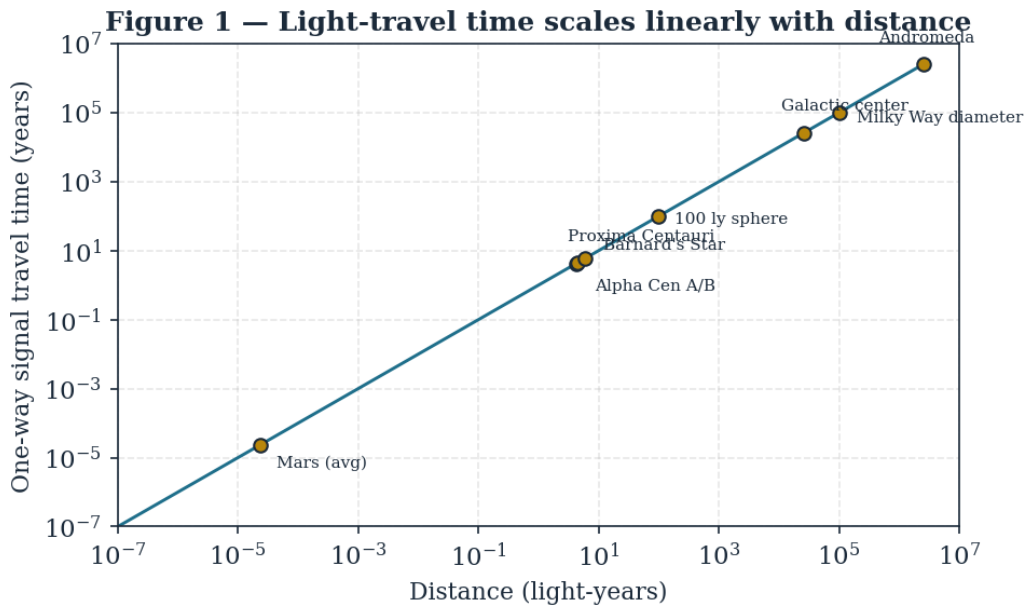


Figure 1. One-way signal travel time versus distance. Nearby stars are reachable in years to decades; the galaxy in $\sim 10^5$ yr; Andromeda in $\sim 2.5 \times 10^6$ yr.

2. The Beacon Strategy: Link-Budget Physics

The most efficient way to broadcast is *not* to radiate isotropically, which wastes essentially all the energy. A circular aperture of diameter D at wavelength λ has gain G , and a transmitter's effective isotropic radiated power is $\text{EIRP} = P_t G$:

$$G = \eta \left(\frac{\pi D}{\lambda} \right)^2, \quad \text{EIRP} = P_t G$$

Arecibo's planetary radar ($D = 305$ m, 1 MW, 2.38 GHz) thereby reached $\text{EIRP} \approx 2 \times 10^{13}$ W (~ 20 TW) in its beam — proof that detectable beacons are within reach of existing technology. The power received at range R by a collector of effective area A_{eff} , and the resulting detection range from the radiometer equation (threshold SNR, channel bandwidth B , integration time τ , system temperature T_{sys}), are:

$$P_r = \frac{\text{EIRP} A_{\text{eff}}}{4\pi R^2}, \quad R_{\text{max}} = \left(\frac{\text{EIRP} A_{\text{eff}}}{4\pi \text{SNR} k T_{\text{sys}}} \right)^{1/2} \left(\frac{\tau}{B} \right)^{1/4}$$

The fourth-root dependence on integration time is the key lever: **patience buys range cheaply**. Taking an Arecibo-class beam detected by a 100 m dish ($A_{\text{eff}} \approx 4.7 \times 10^3 \text{ m}^2$, $T_{\text{sys}} = 25 \text{ K}$, $B = 1 \text{ Hz}$, $\text{SNR} = 10$) gives the ranges in Table 1 and Figure 2.

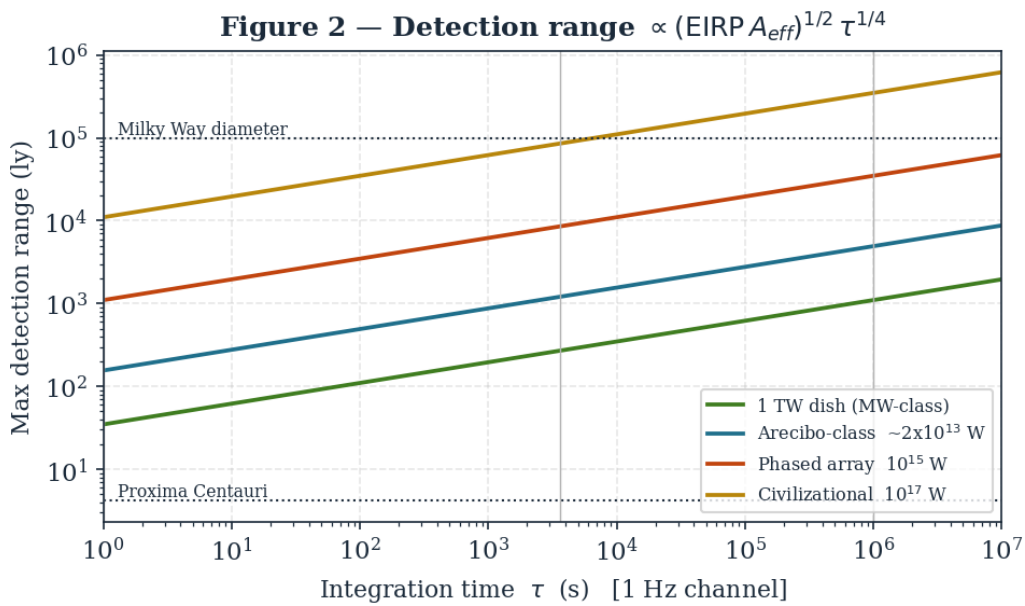


Figure 2. Detection range versus integration time for four EIRP classes. A megawatt dish reaches the nearest stars; galaxy-crossing detection requires phased arrays of $10^{15}\text{--}10^{17} \text{ W}$.

Integration time τ	$(\tau/B)^{1/4}$	Detection range
1 s (snapshot)	1	~155 ly
100 s	3.2	~490 ly
3 hr (10^4 s)	10	~1,550 ly
12 d (10^6 s)	32	~5,000 ly

Table 1. Detection range for an Arecibo-class beacon ($\text{EIRP} = 2 \times 10^{13} \text{ W}$) received by a 100 m dish.

Inverting the range equation gives the transmitter EIRP required to be detectable at a given distance (Figure 3, Table 2). The difficulty scales as R^2 : nearby stars are trivial, while the galactic center is a phased-array undertaking.

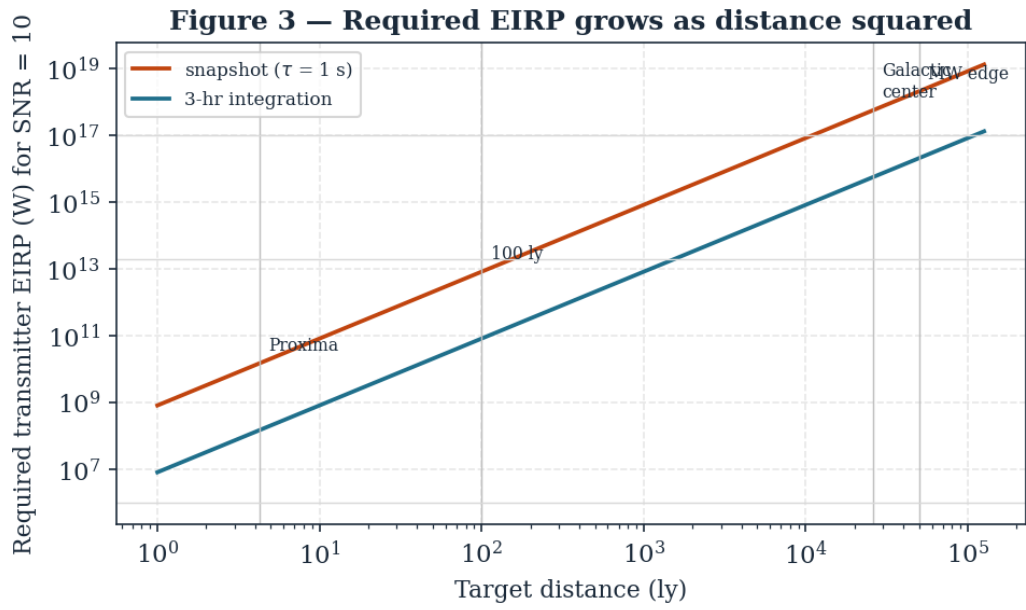


Figure 3. Required EIRP versus target distance for snapshot and 3-hour integration. Nearby stars need megawatt-class transmitters; galaxy-scale reach needs civilizational arrays.

Target	Distance	Required EIRP (snapshot)
Proxima Centauri	4.25 ly	~1.5×10 ¹⁰ W (MW-class on 70 m dish)
100 ly sphere	100 ly	~8×10 ¹² W (Arecibo-class)
Galactic center	~26,000 ly	~5×10 ¹⁷ W (phased array)

Table 2. Required transmitter EIRP for SNR = 10 at a large receiver (1 Hz, snapshot).

3. Beam Geometry, Frequency & Message Design

High gain means a narrow beam. The half-power beamwidth of a circular aperture is $\theta \approx 1.22\lambda/D$. A 70 m dish at 1.5 GHz ($\lambda = 0.2$ m) gives $\theta \approx 0.2^\circ$ (Figure 4), so tiling the whole sky would need $4\pi/\theta^2 \approx 10^6$ pointings — hence one **targets** rather than sweeping blindly. The same beam, however, spans ~940 AU at Proxima, so it comfortably illuminates an entire target system.

$$\theta \approx 1.22 \frac{\lambda}{D}, \quad N_{\text{beams}} = \frac{4\pi}{\theta^2} \approx 10^6$$

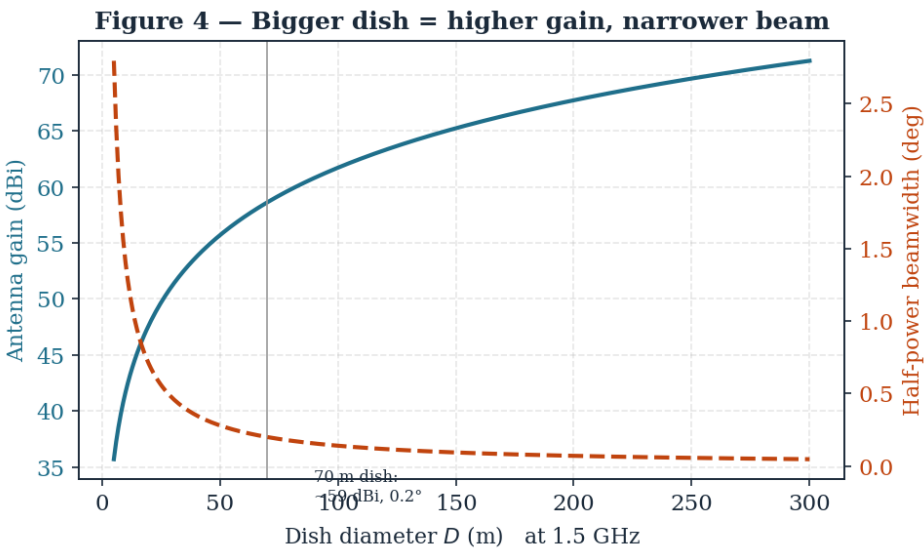


Figure 4. Antenna gain (left axis) and half-power beamwidth (right axis) versus dish diameter at 1.5 GHz. Larger apertures trade sky coverage for reach.

Frequency. The primary band is the ‘water hole’ (~1.42–1.66 GHz), bracketed by the neutral-hydrogen and hydroxyl lines — galactically quiet and an ‘obvious’ meeting place. Secondary bands add pulsed optical/IR lasers (very high gain from small apertures) and wideband data bursts. **Message.** The payload is layered so a recipient can lock on, decode, and ultimately rebuild the network (Table 3).

Layer	Content	Purpose
1 — Artificiality	Primes, hydrogen-line ref, binary counting	Pass the ‘not-natural’ test
2 — Decode key	Grammar, units, time/frequency, coordinates	Make the rest decodable
3 — Archive	Biology, science, Earth location, culture	Who sent this
4 — Replication	Instructions to build another beacon	Turn receivers into transmitters

Table 3. Layered message architecture. Layer 4 is the load-bearing innovation: the goal is not to be heard once but to create more broadcasters.

4. Beacon One: The Buildable Phase-1 Pilot

Beacon One proves we can send a deliberately detectable, repeatable, decodable, self-replicating signal to the nearest ~100–1,000 stars using TRL 8–9 hardware. Three build options scale with ambition (Table 4); the recommended configuration pairs a leased-dish pathfinder with a dedicated dish (Table 5).

Option	Approach	Cost
A — Minimum viable	Lease existing dishes for transmit windows	\$5M – \$25M
B — Dedicated station	30–70 m dish + 100 kW–1 MW TX	\$75M – \$300M

Option	Approach	Cost
C — Phased array	Hundreds–thousands of small dishes	\$300M – \$1.5B

Table 4. Phase-1 build options.

Component	Estimated cost
Scientific & legal planning	\$2M – \$5M
Message / protocol design	\$1M – \$3M
Observatory leasing & testing (pathfinder)	\$5M – \$20M
Custom high-power transmitter + encoder	\$10M – \$40M
Dedicated dish or retrofit (30–70 m)	\$30M – \$100M
Operations, 10 years	\$20M – \$80M
Public archive + governance	\$2M – \$10M
TOTAL	\$70M – \$250M

Table 5. Recommended Beacon One bill of materials. The total sits within the envelope of Breakthrough Listen (\$100M/10 yr) and well below SKA Phase 1 (~€2B).

Targets: Proxima & Alpha Centauri, Barnard’s Star, Wolf 359, Lalande 21185, Sirius, Tau Ceti, Epsilon Eridani, nearby red dwarfs, and confirmed exoplanet systems within 100 ly. **Cadence:** minutes per target, revisited weekly, sustained ≥ 10 years on predictable timing so recipients can integrate across many passes. Confirmation of reception is explicitly not a Phase-1 deliverable — light-travel time alone forecloses it.

5. The Solar Gravitational Lens Relay

To raise link strength one can add power, build a bigger antenna, or **borrow** one. A ray grazing the solar limb is deflected by the Einstein angle α , focusing rays onto the axis at a minimum distance $d_{\min}$:

$$\alpha = \frac{4GM_{\odot}}{c^2 R_{\odot}} \approx 1.75'', \quad d_{\min} = \frac{R_{\odot}}{\alpha} \approx 549 \text{ AU}$$

Beyond ~550 AU (~14× Pluto’s orbit, ~3.3× Voyager 1’s current distance) the spacecraft stays in focus along a half-line. The lens concentrates flux into an Einstein ring, yielding effective gain of order 10^{11} at the 21 cm line — equivalent to a ~25 km dish — and far more at optical wavelengths (Figure 5):

$$G_{\text{SGL}} \sim 4\pi^2 \frac{r_g}{\lambda}, \quad r_g = \frac{2GM_{\odot}}{c^2} \approx 2.95 \text{ km}$$

Figure 5 — The Sun-as-antenna gain rises toward short wavelengths

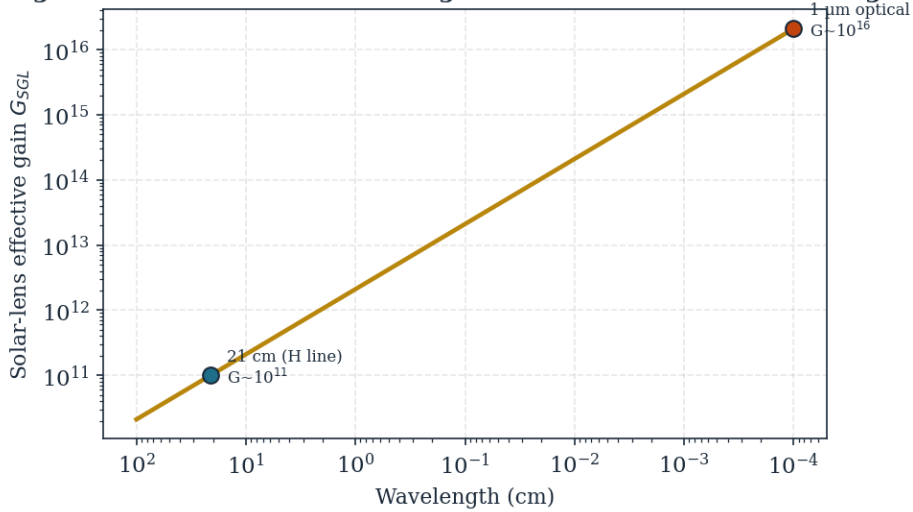


Figure 5. Solar-lens effective gain rises toward short wavelengths ($G \propto 1/\lambda$). At 21 cm the gain is $\sim 10^{11}$ (~25 km equivalent aperture); in the optical it is orders of magnitude higher.

The catch is geometric: enormous gain, but along one axis at a time, so reaching N targets needs N probes. Getting to the focus is the hard part (Table 6); planets make poor lenses because focal distance scales inversely with mass (Table 7).

Propulsion	Speed	Time to 550 AU
Voyager-class (chemical + assists)	~3.5 AU/yr	~150 yr
Solar-sail Oberth (sundiver)	~20–30 AU/yr	~20–50 yr
Future nuclear-electric / sail	>40 AU/yr	<15 yr

Table 6. Transit time to the solar focus. Per-probe cost ~\$1B–\$5B (probe + sundiver stage + launch + decades of operations).

Body	Focal distance	Useful as lens?
Sun	~550 AU	Yes — strong, clean focus
Jupiter	~6,000+ AU	Mostly no — too distant
Planetary atmospheres	—	No — noisy, unstable, hard to aim

Table 7. Why the Sun is the only practical Solar-System lens. The SGL is a Phase 3–4 relay amplifier, not a broadcaster.

6. The Orchard Network: Self-Replicating Deep-Time Infrastructure

The end-state replaces transmitted messages with physical, persistent, reproductive **seeds** (10 kg–1 ton) that cruise, lie dormant, self-repair, replicate from local material, and invite emerging intelligences to join. If seeds hop between systems separated by d at speed v , dwelling T_{dwell} to replicate, the colonization wavefront and the population growth are:

$$V_{\text{wave}} = \frac{d}{d/v + T_{\text{dwell}}}, \quad N(t) = N_0 k^{t/T_{\text{gen}}}$$

With $d \approx 5$ ly and $T_{\text{dwell}} \approx 10^3$ yr, the galaxy fills in $\sim 10^7$ – 10^8 yr — and crucially, beyond $\sim 5\%$ c the wavefront is limited by *replication time*, not *cruise speed* (Figure 6). Survival over deep time follows reliability theory; a single node is fragile, but N independent nodes make the network effectively immortal (Figure 7):

$$s(t) = e^{-pt}, \quad S_{\text{net}} = 1 - (1 - s)^N$$

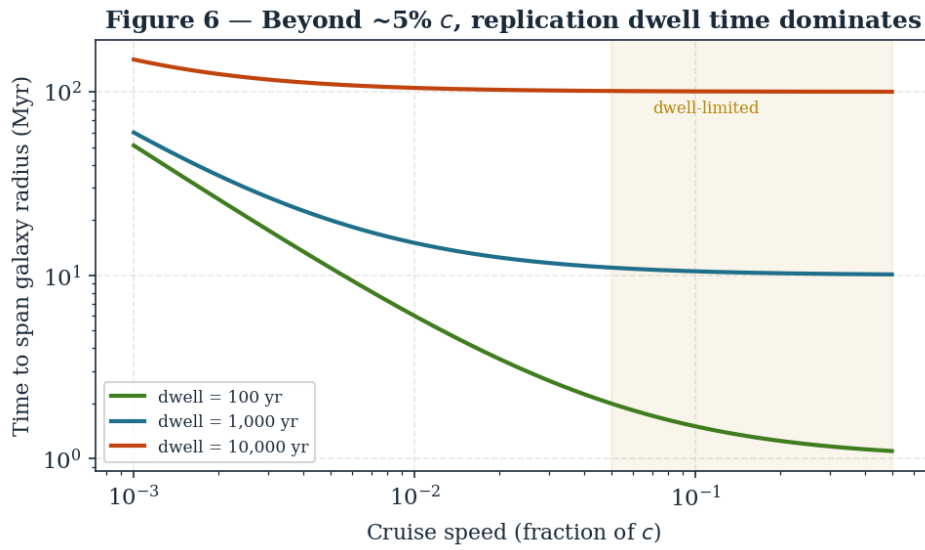


Figure 6. Time to span the galaxy versus cruise speed, for three replication dwell times. The curves flatten past $\sim 5\%$ c : how fast a seed builds matters more than how fast it flies.

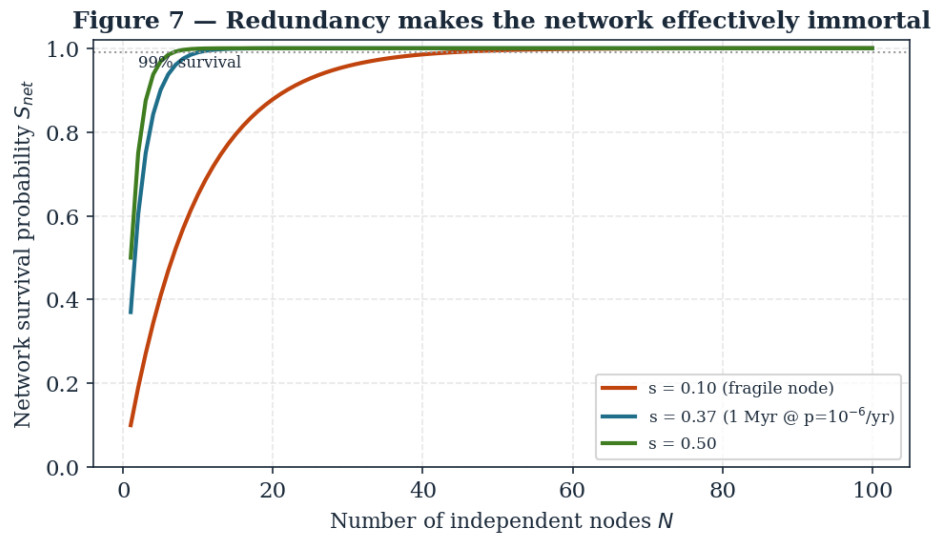


Figure 7. Network survival probability versus number of independent nodes. Even fragile nodes yield a near-certain network with modest redundancy — replication is the survival mechanism.

Nodes N	s = 0.37	s = 0.10
1	0.37	0.10
10	0.99	0.65
100	>0.999999	0.99997

Table 8. Network survival $S = 1-(1-s)^N$. Access to the seed archive is gated by layered disclosure (math → science → engineering → culture → unrestricted) so a young civilization is never overwhelmed.

7. Program Roadmap & Cost

The five phases overlap and each is independently valuable; Phase-1 hardware and the message standard are reused throughout. Costs span five orders of magnitude (Figure 8, Table 9), from a fundable pilot to a civilizational endeavor.

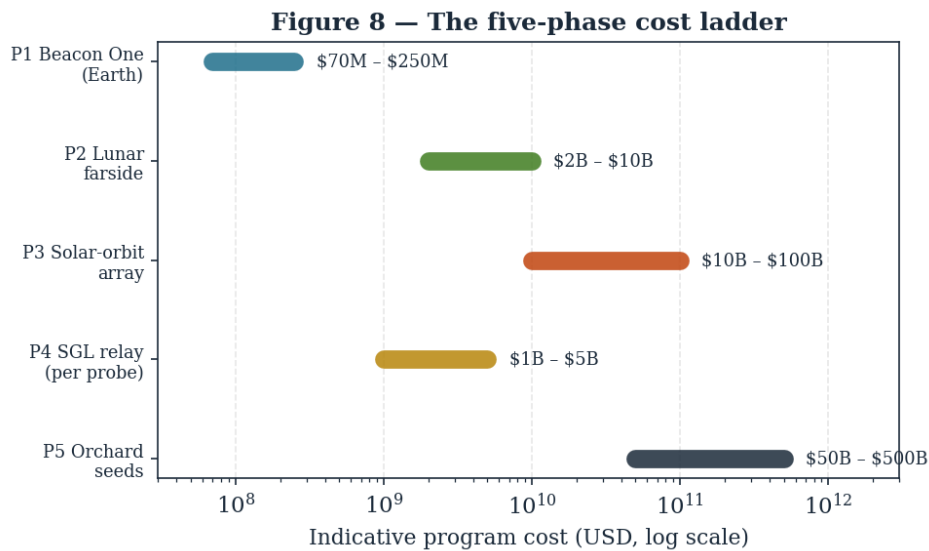


Figure 8. The five-phase cost ladder (log scale). Phase 1 is comparable to a single science mission; later phases scale toward civilizational investment.

Phase	System	Horizon	Indicative cost	Primary capability
1	Beacon One (Earth)	0–10 yr	\$70M–\$250M	Targeted beams, ~10 ³ stars
2	Lunar farside array	10–30 yr	\$2B–\$10B	Radio-quiet large aperture
3	Solar-orbit array	30–80 yr	\$10B–\$100B	All-sky access, plane sweeps
4	SGL relay (>550 AU)	50–150 yr	\$1B–\$5B/probe	Ultra-high-gain links
5	Orchard seeds	10 ² –10 ⁶ yr	Civilizational	Self-replication, archive

Table 9. Phased roadmap. Beacons (Phases 1–3) find and invite; SGL relays (Phase 4) carry deep conversation along proven axes; Orchard seeds (Phase 5) make the network self-sustaining.

8. Conclusion

Detection range grows as the square root of EIRP and receiver area and only the fourth root of integration time, so directed, repeated, multi-band beams at selected targets dominate isotropic broadcasting on every axis that matters. The nearest stars are reachable now with a \$70M–\$250M Earth beacon; the Sun offers a ~25-km antenna to any probe placed beyond 550 AU; and self-replicating seeds can populate the galaxy in a cosmic eyeblink while surviving deep time through redundancy. The architecture’s deepest purpose is not contact but continuity.

“We are one civilization in one era. We may not survive to hear you; you may not survive to hear others. But if you receive this — build another beacon, add your archive, and send it onward. Let intelligence become cumulative across the dark.”

References & Benchmarks

Friis transmission and radiometer equations (standard). Arecibo S-band planetary radar (~2×10¹³ W EIRP). Water-hole rationale: Oliver & Billingham, *Project Cyclops* (1971). Target selection: Breakthrough Listen catalogs; program funding benchmark \$100M/10 yr. Facility costs: SKA Phase 1 (~€2B), FAST (~\$170M), Allen Telescope Array (~\$50M partial). Solar gravitational lens: Eshleman,

Science (1979); C. Maccone, *Deep Space Flight and Communications*; JPL solar-sail SGL mission concepts (~25-yr transit). Voyager 1 trajectory: >160 AU, ~3.5 AU/yr. Self-replicating-probe galaxy-filling timescales: von Neumann probes; Tipler (1980); Fermi-paradox literature (~ 10^6 – 10^8 yr). All cost and physics figures are order-of-magnitude engineering estimates for planning, not quotations.